# Infinite Time Does Not Make Every Observer Inevitable

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## Abstract

An endless sequence need not contain every possible event: the even numbers never produce an odd number. The same distinction matters for Boltzmann brains, proposed observers formed by rare fluctuations instead of biological development. A cosmological model must show that such a system can form, that it can function as an observer, and how much weight its observations receive. Infinite time alone settles none of these questions. Determinism alone does not exclude fluctuation observers either. This paper separates those requirements and shows why they must be checked in the particular physical model.

## 1. Introduction

A Boltzmann brain is a proposed observer produced by a fluctuation rather than the usual history of biological development. Some cosmologies give such observers enough weight to threaten the reliability of the records used to infer the cosmology. The inference depends on formation dynamics, what counts as an observer and how observations are weighted. Infinite time is only one input. Coherent memories alone cannot exclude a fluctuation stipulated to contain those same memories; the issue is what the model predicts about evidence and its reliability (Carroll, 2017).

Existing responses include vacuum decay, stationary vacuum accounts, models with asymptotic freezing, and different cosmological measures. Boddy, Carroll and Pollack (2015) distinguish a projection associated with a putative observer from a process that forms and operates one. Tumulka (2018) studies freezing in a specified Bohmian field model. Both already question the relevant formation process. The present discussion builds on that work and does not claim that the literature ignored dynamics.

The useful question is narrower than whether every possible microstate appears. Does the specified cosmology realize histories with the organization attributed to a functioning observer, and with what weight? The following sections separate this question from ergodicity, branch language and algorithmic description length.

## 2. Three requirements behind a dominance claim

A fluctuation observer argument can be organized as follows. Each step requires a model; none follows from the word infinite.

(P1) The cosmology supplies an appropriate duration and volume, together with a law for transitions or formation events.

(P2) Under that law and its boundary conditions, a specified class of candidate observer histories has positive occurrence weight or rate.

(P3) Those histories meet the adopted criterion for experience and records.

(C) Under a specified weighting and conditioning rule, their observations dominate those of observers with ordinary developmental histories.

The conclusion requires comparing observer weights, not merely establishing a nonzero rate. If both candidate populations are infinite, a regulator or other measure prescription is needed. A result that changes with that prescription is not determined by infinity alone.

For a probability preserving map T on (X, ), ergodicity means invariant measurable sets have measure zero or one. The ergodic theorem then gives the long run visit frequency (B) for a measurable target B, for almost every starting point, when T is ergodic. This concerns positive measure sets and typical starts. It does not assert that every exact point is visited, that every starting point behaves typically, or that B constitutes an observer. A formation model can also give positive rates without global ergodicity.

## 3. Determinism does not settle reachability

### 3.1. Unitary evolution and branch structure

In an Everettian formulation, the quantum state evolves unitarily. Which states it reaches depends on the Hamiltonian and initial condition. Decoherent histories supply a framework for discussing records and branches; the mere choice of a basis vector does not define a branch containing an observer.

Unitary evolution can turn a zero coefficient in a chosen basis into a nonzero coefficient. In units with =1, take H= x and the initial state |0 . Then | (t) =cos( t)|0 −i sin( t)|1 . The probability of |1 begins at zero and reaches one at t= /(2 ). What remains unoccupied is an invariant subspace initially assigned zero weight, not every basis direction with an initially zero amplitude.

A Hamiltonian diagonal in its energy basis preserves the magnitudes of the corresponding energy coefficients. That restricted fact cannot be transferred to an arbitrary configuration basis. To exclude a candidate observer history, establish an invariant restriction, a vanishing relevant amplitude throughout the evolution, or another dynamical obstruction. None is supplied by determinism alone.

A nonzero projection at an instant likewise does not by itself establish an operating observer. The physical claim concerns the organization and evolution attributed to the projected system. This distinction leaves a substantive question for each cosmology rather than treating every component of a mathematical expansion as a completed conscious history.

### 3.2. Bohmian mechanics

Tumulka's concrete cosmological model uses a Bohmian scalar field on a de Sitter background. Its late evolution becomes too simple to support the functioning attributed to a brain. The argument concerns field dynamics and a specified class of states. It should not be summarized as a theorem that every particle configuration in every Bohmian cosmology freezes.

Conserved quantities can restrict trajectories to invariant sets. Ergodicity must then be assessed relative to a specified invariant measure on the appropriate set, such as an energy surface. Confinement within that set does not refute ergodicity there. Nor does the presence of an attractor follow from determinism; Hamiltonian and dissipative systems require different analyses.

### 3.3. Deterministic examples

A finite counter gives the simplest contrast. Let X={0,…,N−1}, T(x)=x+1 mod N, with the uniform measure. Every orbit visits every state, and the only invariant subsets are empty and all of X. This deterministic discrete system is ergodic. On the same X, T(x)=x leaves every state fixed and is not ergodic when N>1. The rule, rather than determinism, decides the result.

An irrational rotation x x+  mod 1 gives a familiar infinite mathematical example. It is ergodic for uniform length measure, although a single orbit is countable and cannot contain every point of the circle. Density and appropriate visit frequencies do not mean literal exhaustion of the continuum. This is a model used to clarify the mathematics, not a commitment to a physical continuum. See Green's lecture notes, section 3.4.

## 4. Infinite does not mean everything

The phrase "given infinite time, every possible configuration will be realized" trades on an ambiguity. The set of configurations visited by a deterministic trajectory over infinite time can be infinite without being exhaustive. This is a mathematical triviality: the even integers are an infinite subset of the natural numbers that contains no odd integers. An infinite decimal expansion of a rational number like $1/7 = 0.142857142857\ldots$ repeats forever without ever producing the digit sequence "999."

The examples show why an infinite orbit and an ergodic orbit must be distinguished. They also show why excluding one exact microstate is insufficient:

an observer criterion normally selects a class of admissible configurations or histories. Its measure and dynamical accessibility are the relevant quantities.

Suppose a proposed formation process is a Poisson process with constant rate $>0$ in the chosen region. The probability of no event by time T is $\exp(-\ T)$, which tends to zero as T increases without bound. That model gives almost sure eventual formation without a separate proof of global cosmological ergodicity. The substantive assumptions are the positive rate and the validity of the process over the interval.

A time dependent rate can give a different answer. For $(t)= \exp(-t/\ )$, the integrated rate over infinite time is , and the probability of at least one event is $1-\exp(-\ )$, below one for finite . Infinite time did not make the integrated opportunity infinite. Neither illustrative rate is asserted to describe our universe. Recurrence theorems likewise have hypotheses, including appropriate spectral or finite measure conditions, and recurrence alone establishes no observer dominance.

## 5. Mathematical randomness and description length

Matheism treats regularity and randomness as mathematical. This does not collapse stochastic laws into pseudorandom algorithms. A probability measure is a mathematical structure even when it assigns probability one to a class of individual infinite sequences lacking finite generating programs. The ontology does not select a cosmological transition law by itself.

A deterministic generator with finitely many internal states eventually repeats. That result does not apply to every deterministic computation with unbounded memory. Conversely, for independent fair coin trials every specified finite binary word appears with probability one. Statements about randomness must name the distribution and event. Being a fixed mathematical sequence does not prevent the sequence from having the required frequencies.

### 5.1. Not all random distributions produce all configurations

A distribution's support matters. A process confined to [0,1] cannot output 2. That example states a restriction already built into the model. It offers no restriction on brain histories until the physical model identifies the corresponding allowed set.

Small weight is different from zero weight. A nondegenerate Gaussian has positive density throughout its Euclidean domain, although distant regions receive little probability. One cannot exclude those regions by calling their weights exponentially small. Whether repeated opportunities overwhelm a small rate is the integrated rate calculation, not a consequence of the adjective random.

For a projector P and state $|\ (t)\ \rangle$, the Born probability of the associated measurement outcome is $\langle\ (t)|P|\ (t)\ \rangle$. Relating this probability to the formation and operation of an observer requires a model of histories and records. A stochastic

formation rate, a quantum projection probability and an algorithmic prior are distinct objects.

A proposed solution should therefore specify which requirement it changes: availability of suitable histories, their capacity to support experience, or their weight relative to other observations. The existing literature supplies different accounts of these requirements. Their conclusions cannot be combined without their assumptions.

A simplicity comparison belongs at the third stage and requires equal treatment of the candidates.

A developed observer may have a compact generative description in terms of laws, boundary conditions, time and location. A fluctuation observer may also be generated by a compact cosmological model. It is invalid to charge the first candidate a generator while forcing the second to list every constituent independently.

In a deterministic computable model, a specified event e extracted from a finite history obeys K(e) K(L)+K(I)+K(t)+K(j)+O(1), where j identifies the relevant region or extraction instruction. Very late times and complicated locations can carry cost. In a stochastic model, the likelihood of the event also matters; a compact sampling rule is not a compact code for every individual outcome.

Even an established description length difference $\Delta$L gives a factor $2\hat{}\Delta$L only under the stated code prior. Posterior observer odds also require likelihoods and a rule for weighting observer locations. No general suppression factor for Boltzmann brains has been calculated here.

### 5.2. The scope of a sufficient reason argument

A strong Principle of Sufficient Reason may motivate seeking deeper explanations of outcomes. Whether a probabilistic law supplies such a reason is a philosophical dispute. Neither accepting that principle nor rejecting fundamental chance proves nonergodicity. The two deterministic maps in section 3.3 already settle that logical point. A metaphysical commitment must be connected to a specific dynamics before it answers the formation question.

## 6. Comparison with existing approaches

Two existing responses illustrate the difference between specifying a dynamics and simply labeling it deterministic.

Boddy, Carroll and Pollack's stationary vacuum proposal depends on assumptions about the state and its larger Hilbert space. Their earlier analysis also distinguishes finite closed de Sitter systems, where recurrences and dynamical fluctuations remain possible. Their conditional conclusions supply no blanket exclusion under unitary dynamics (2014; 2015).

Tumulka's conclusion depends on the field model and its late dynamics. His paper explicitly distinguishes configurations shaped like brains from configurations capable of functioning as brains. This is an example of the organizational requirement, not proof that a small instantaneous Born probability vanishes.

The present framework organizes those dependencies. It supplies elementary counterexamples to invalid general inferences and identifies the additional calculation needed in a proposed cosmology. It does not replace the existing models with a stronger exclusion theorem.

## 7. Objections and replies

**Objection 1: Does allowing stochastic laws undermine the structural account?**

Reply: No. Both deterministic maps and probability measures have mathematical structure. The claim that reality is mathematical does not decide between them without further premises. The present argument tests formation and observer weighting under whichever law the model specifies.

**Objection 2: Even if the exact Boltzmann brain configuration is never visited, configurations "close enough" to constitute a brain will be visited.**

Reply: Approximate functioning is the appropriate issue. Specify a class of histories robust enough to implement the relevant organization. Ergodicity can imply repeated visits to a positive measure target under its hypotheses, while a rate calculation may establish formation by another route. Nonergodicity alone cannot show that a particular target is avoided.

**Objection 3: Does ordinary thermodynamic behavior require visiting every exact microstate?**

Reply: No. Equilibration of selected observables is different from exhaustive visitation. A deterministic theory can reproduce thermodynamic regularities under suitable conditions. The low entropy boundary and the formation of reliable records remain separate issues; they are not consequences of either the word deterministic or the word ergodic.

**Objection 4: In Many-Worlds, every branch exists, so every configuration with non-zero amplitude is realized somewhere.**

Reply: A branch is an account of correlated records and dynamics, not an arbitrary nonzero coefficient in any chosen basis. Once a model establishes suitable observer histories, compare their weights using the same observation rule used for ordinary observers. A simplicity prior cannot silently replace the Born rule or a cosmological measure.

## 8. Implications

The framework gives three concrete tasks for a proposed cosmological account.

First, define the target as a class of functioning histories and calculate reachability or formation rates under the proposed state and dynamics. State whether the result holds for every allowed initial condition or only almost surely under a specified measure.

Second, separate formation from dominance. Integrate rates over the relevant duration and volume, compare ordinary observer production and disclose any regulator needed for infinite populations. A small nonzero formation rate and a large observer count are not yet a posterior probability for our evidence.

Third, condition on the evidence consistently. Wolpert, Rovelli and Scharnhorst (2025) distinguish the roles of assumptions about memories, temporal conditioning and entropy. A model should state which records it treats as data and which as reliable traces of a past. No title or slogan resolves that choice.

## 9. Conclusion

Infinite time alone does not guarantee a fluctuation observer. Determinism alone does not exclude one. The model must specify which histories are possible, which systems function as observers and how their observations are weighted.

The even integers go on forever without producing an odd integer. A cyclic counter visits every state. Which analogy applies to a universe is a question for its dynamics.

## References


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